

B.Tech. Degree V Semester Special Supplementary Examination

July 2016

ME 1506 POWER PLANT ENGINEERING

(2012 Scheme)

Time : 3 Hours

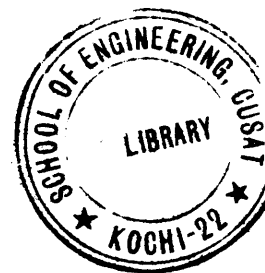
Maximum Marks : 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) What is the relevance of local factor on different types of consumers?
- (b) What are the suitable site conditions for a Buttress Dam?
- (c) What are the essential components of diesel power plants?
- (d) List out the advantages of closed cycle gas turbine plants.
- (e) What are the different methods for governing of a steam power plant?
- (f) Describe on environmental aspects of nuclear power plants.
- (g) Enumerate three main ways to exploit solar energy.
- (h) What are the advantages of fuel cell over other non-conventional power systems?



PART B

(4 × 15 = 60)

- II. (a) Describe the difficulties to meet peak load. (5)
 - (b) A small plant capital cost is ₹50,000/- and its salvage value is ₹5000/- at the end of a useful life of 20 years. Find its value half way through its life based on: (10)
 - (i) Straight line depreciation method.
 - (ii) Sinking fund depreciation at 5% compounded annually.
- OR**
- III. (a) What are the various costs in economics of power plants? (5)
 - (b) Describe various types of spillways and conditions applicable for each one. (10)
- IV. Answer the following: (15)
 - (i) Why is supercharging necessary in diesel power plants?
 - (ii) List out the classification of gas turbine plants according to different criteria.
 - (iii) Brief out the working of the governing system of gas turbine power plants.
- OR**
- V. (a) What are the advantages and disadvantages of diesel plants over thermal plants? (10)
 - (b) What are the factors considered for selection of underground fuel storage? (5)
- VI. (a) What are fission and fusion processes? Compare them. (8)
 - (b) What are the major advantage of FBC system over the conventional one. (7)
- OR**
- VII. (a) Compare fire tube and water tube boilers. (8)
 - (b) Differentiate travelling gate stoker firing and underfeed stoker firing. (7)
- VIII. Write notes on the following. (15)
 - (i) Batch type biogas plant.
 - (ii) What are the advantages of fuel cell power generation?
 - (iii) Factors affecting the suitability of the site for tidal power plant.
- OR**
- IX. Write notes on the following: (15)
 - (i) Working of Thermionic conversion system.
 - (ii) What are the disadvantages of wind power?
 - (iii) Different types of concentrating type collectors.